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Original Article

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A digital, real-time, history-based sleep-management tool to enhance alertness

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Abstract

In today's 24-h society, chronic sleep disruption and circadian misalignment have led to a "global sleep crisis," increasing the risk of cognitive impairment, workplace accidents, and long-term health consequences. Yet, most sleep management strategies rely on one-size-fits-all recommendations that overlook individual variability, resulting in suboptimal and/or impractical solutions. To address this, we previously developed a real-time, personalized sleep scheduling framework based on tracking of dynamic sleep pressure and circadian rhythms using a mathematical model. We recently implemented this framework in SleepWake, a mobile app designed for real-world applications. In a retrospective analysis of a 71-participant clinical study and a prospective trial with 19 shift workers, greater adherence to SleepWake's personalized recommendations led to significant improvements in alertness. These benefits stemmed from two key innovations that go beyond static sleep guidelines: real-time prescribed supplemental sleep to counteract prior deficits and personalized sleep phase alignment tailored to individual circadian patterns. This study provides the first direct evidence that continuously updated, individualized sleep schedules can optimize alertness and sleep health in real-world settings. By delivering sleep recommendations rooted in sleep physiology and evidence-based modeling, SleepWake has the potential to improve health, enhance safety, and elevate the overall quality of life in today's around-the-clock society.

Key words: sleep; circadian rhythm; mathematical model; mobile application; shift work; alertness; sleep intervention

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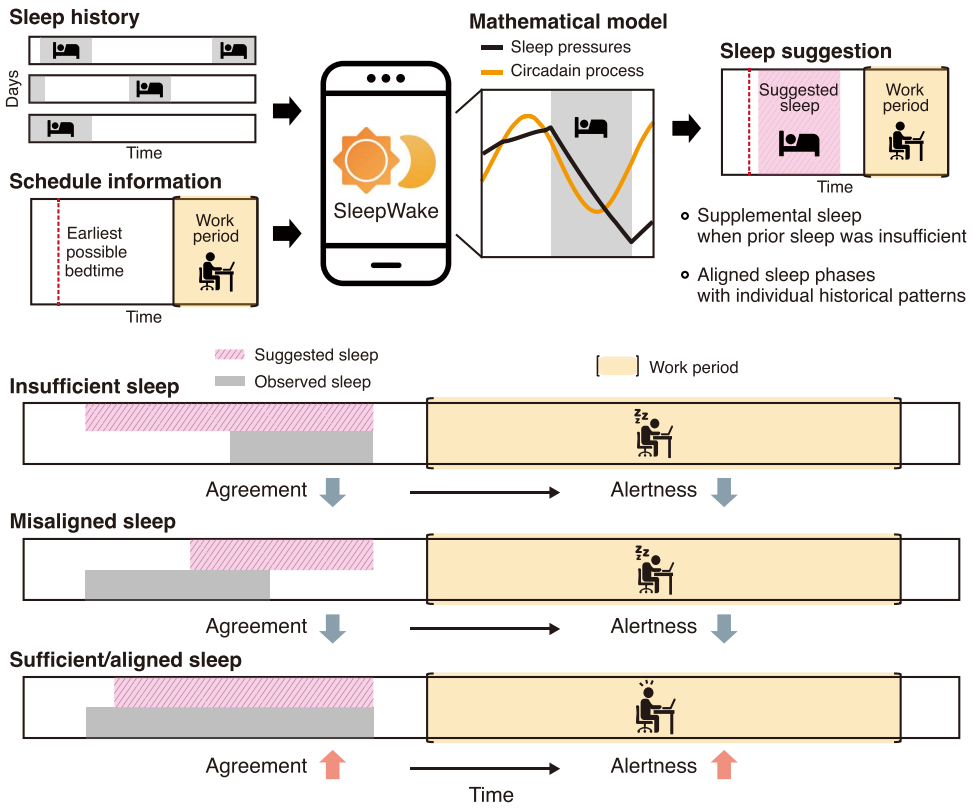
Graphical Abstract



Statement of Significance

The increasing demand for work outside daylight hours poses significant health and safety risks. To address these challenges, we previously developed a real-time, personalized sleep scheduling framework powered by a mathematical model. In this study, we implemented the framework through SleepWake, a mobile app that tailors sleep schedules based on real-time updates of individual sleep history. Unlike generic guidelines, SleepWake dynamically adjusts sleep schedules, offering additional rest when needed and aligning sleep with natural rhythms. Among 71 female and 19 male shift workers, those who followed the app's recommendations showed significant improvements in alertness. These findings underscore the benefits of continuously updated, personalized sleep schedules grounded in sleep physiology and evidence-based modeling for optimizing alertness.

Introduction

Artificial light is pervasive in modern society, and nighttime activities have become both common and crucial for maintaining essential societal functions [1, 2]. In Europe, approximately 20 per cent of workers engage in night shifts at least once a month [3], with a higher rate in the United States [4]. In South Korea and Japan, over 10 per cent of workers participate in night shifts [5, 6]. Furthermore, it has been suggested that over 80 per cent of the general population may lead lifestyles that resemble those of shift workers [7], highlighting the widespread prevalence of nighttime activities that extend beyond traditional shift work. This trend is further supported by a recent study reporting that high-GDP countries tend to have later bedtimes [8].

Widespread nocturnal activity disrupts natural sleep patterns and circadian rhythms, which are inherently synchronized with the light–dark cycle [9, 10]. These disruptions result in sleep deprivation and poor sleep quality, which can have profound impacts on individual health. These impacts include immunological and metabolic changes and an increased risk of diseases, including diabetes and certain types of cancer [11–14]. Moreover, insufficient sleep and misaligned work schedules impair alertness and cognitive function, reducing productivity and elevating the risk of errors, injuries, and accidents [15–17]. These issues present significant challenges to public health and safety and the overall well-being of society.

In response, a variety of nonpharmacological interventions, such as improvements in sleep hygiene and light therapy, have been developed [18–20]. These interventions aim to promote sufficient, biologically appropriate sleep and optimize alertness when required, even in contexts where biological rhythms are misaligned with societal demands. A key strategy in these interventions is the establishment of appropriate sleep schedules [21]. However, most existing approaches rely on one-size-fits-all sleep schedules, leading to conflicting recommendations: some studies advocate for morning sleep after a night shift [22], while others suggest evening sleep [23–25], split sleep [26], or brief naps [27]. These approaches also fail to account for individual differences in factors such as sleep history and work schedules. Thus, they are often impractical and do not accommodate the diverse sleep patterns required to meet individual needs and circumstances [28]. Furthermore, even within the same individual, optimal sleep patterns may vary depending on factors such as prior sleep history.

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To overcome the shortcomings of these one-size-fits-all sleep interventions, recent advancements in mathematical modeling have paved the way for personalized sleep strategies [29–33]. These models leverage individual sleep histories to predict real-time fluctuations in sleep pressure and circadian rhythms, the primary regulators of sleep and alertness [29, 34–36]. Based on these predictions, the framework generates physiologically optimal sleep schedules tailored to enhance alertness during individual-selected periods, unlike previous uniform approaches [29–33]. This adaptability is particularly beneficial for individuals with irregular or highly variable schedules, offering practical solutions to the diverse and dynamic nature of sleep needs. Furthermore, the integration of these frameworks into real-world settings has been accelerated by the widespread use of digital technologies, including wearable devices and smartphones equipped with advanced sensors and algorithms [37, 38].

These interventions incorporating model-based sleep scheduling can improve shift workers' sleep by increasing total sleep time (TST), reducing sleep onset latency, and alleviating insomnia [30–32]. However, some of the previous studies combined sleep scheduling with additional educational resources—such as guidance on caffeine use and light exposure—making it difficult to isolate the effects of sleep schedule adjustments [30, 31]. Importantly, prior intervention studies utilizing mathematical models lacked detailed analyses of how participants' sleep patterns changed in response to these interventions [30–32]. As a result, it remains unclear whether the observed benefits stemmed directly from sleep schedule modifications or were influenced by external factors. Additionally, these studies relied on estimated sleep data based on activity schedules rather than observed sleep histories to simulate the models, potentially compromising the accuracy of alertness predictions [29]. This limitation may explain why one study failed to observe a clear improvement in alertness despite implementing a model-based intervention [31].

This study aimed to address a previously unresolved question: does adherence to continuously updated sleep schedules—tailored through a mathematical model integrating sleep history—enhance alertness? We recently developed a framework that generates personalized and effective sleep schedules to optimize alertness for upcoming work periods while maintaining physiologically feasible sleep patterns based on predictions from a mathematical model [29]. Our model leverages real-time updates of observed sleep data to forecast alertness, unlike previous intervention frameworks that relied on estimated sleep data derived from activity schedules [30–32]. To apply this framework in a real-world setting, we implemented it as a mobile app, SleepWake. The effectiveness of SleepWake's sleep recommendations was evaluated through a retrospective analysis of a prior clinical study involving 71 female shift workers, followed by a clinical study with 19 male shift workers. By analyzing daily real-time sleep data, we assessed how precise alignment with personalized sleep recommendations and deviations from baseline sleep patterns influenced alertness. Lastly, we analyzed how our personalized intervention integrates previous sleep history to improve alertness—distinguishing it from conventional one-size-fits-all approaches.

Materials and Methods

Overview of the SleepWake app and evaluation of its effectiveness

In our recent study, we introduced a real-time sleep intervention framework based on a physiologically based mathematical model [29]. This model predicts real-time alertness by simulating the dynamic interplay between sleep pressure and circadian processes based on previous sleep history (see *Mathematical model and sleep recommendation* section for details) [29, 34]. Based on these predictions and individual schedules, the framework recommends an optimized sleep pattern called Adaptive Circadian Split Sleep (ACSS) [29]. ACSS is designed to maintain high model-predicted alertness during desired activity periods while being practical and feasible for real-world implementation (see *Mathematical model and sleep recommendation* section for details).

Implementing the ACSS pattern in real-world settings requires real-time sleep data collection and model-based simulations. To support this, we previously proposed the concept of a mobile sleep recommendation app based on ACSS, though it was not implemented at the time. In the present study, we developed SleepWake, a mobile app designed to deliver ACSS-based recommendations (Figure 1, A–C). SleepWake integrates past sleep history (Figure 1, A, left) through Apple Health and Samsung Health, which provide accurate sleep records from smartwatches [39–42]. In cases where sleep data is missing, such as when smartwatches are not worn, users can manually input their sleep records. Additionally, users can input information on future bedtime availability and work schedules (Figure 1, A, right).

Using this input, the app provides real-time alertness predictions around the current time (Figure 1, B, left) and recommends an optimal sleep period (Figure 1, B, right) for the following day based on the integrated model (see *Mathematical model and sleep recommendation* section for details). The recommended sleep period follows the ACSS pattern, starting from the earliest feasible bedtime and continuing until the next day's work period begins. The sleep period represents the minimum sleep required to maintain the average predicted alertness above zero during the upcoming work period while avoiding physiologically infeasible sleep windows.

Additionally, the app collects real-time subjective ratings (Figure 1, C), including daily alertness (1–10 scale) and sleep quality (1–5 scale). The daily alertness item, available in the afternoon each day, prompts participants to reflect on their overall alertness throughout the day, rather than at a specific moment. The prompt was: "How alert did you feel throughout the day?" (translated into Korean). The sleep quality item assesses the perceived quality of the most recent sleep period, using the prompt: "How was your sleep quality last night?" (translated into Korean).

While users are recommended sleep periods by SleepWake (Figure 1, B, right), their adherence to these recommendations varies, potentially influencing the effectiveness of the app. To quantify adherence, we computed an agreement between the observed (Figure 1, D, shaded region) and recommended (Figure 1, D, hatched region) sleep periods as follows:

$$\text{Agreement} = \begin{cases} \dfrac{\textit{Length}\left((\text{Observed Sleep}) \cap (\text{Recommended Sleep})\right)}{\textit{Length}\,(\text{Recommended Sleep})} \\ \quad \text{if Recommended Sleep} \not\subseteq \text{Observed Sleep} \\ \dfrac{\textit{Length}\,(\text{Observed Sleep})}{\textit{Length}\,(\text{Recommended Sleep})} \\ \quad \text{if Recommended Sleep} \subseteq \text{Observed Sleep} \end{cases}, \quad (1)$$

If users did not fully adhere to the recommendation, the agreement reflects the proportion of the recommended sleep period achieved by the observed sleep period (Figure 1, D, top). In cases of partial overlap, only the overlapping portion of sleep is included in the agreement calculation (Figure 1,

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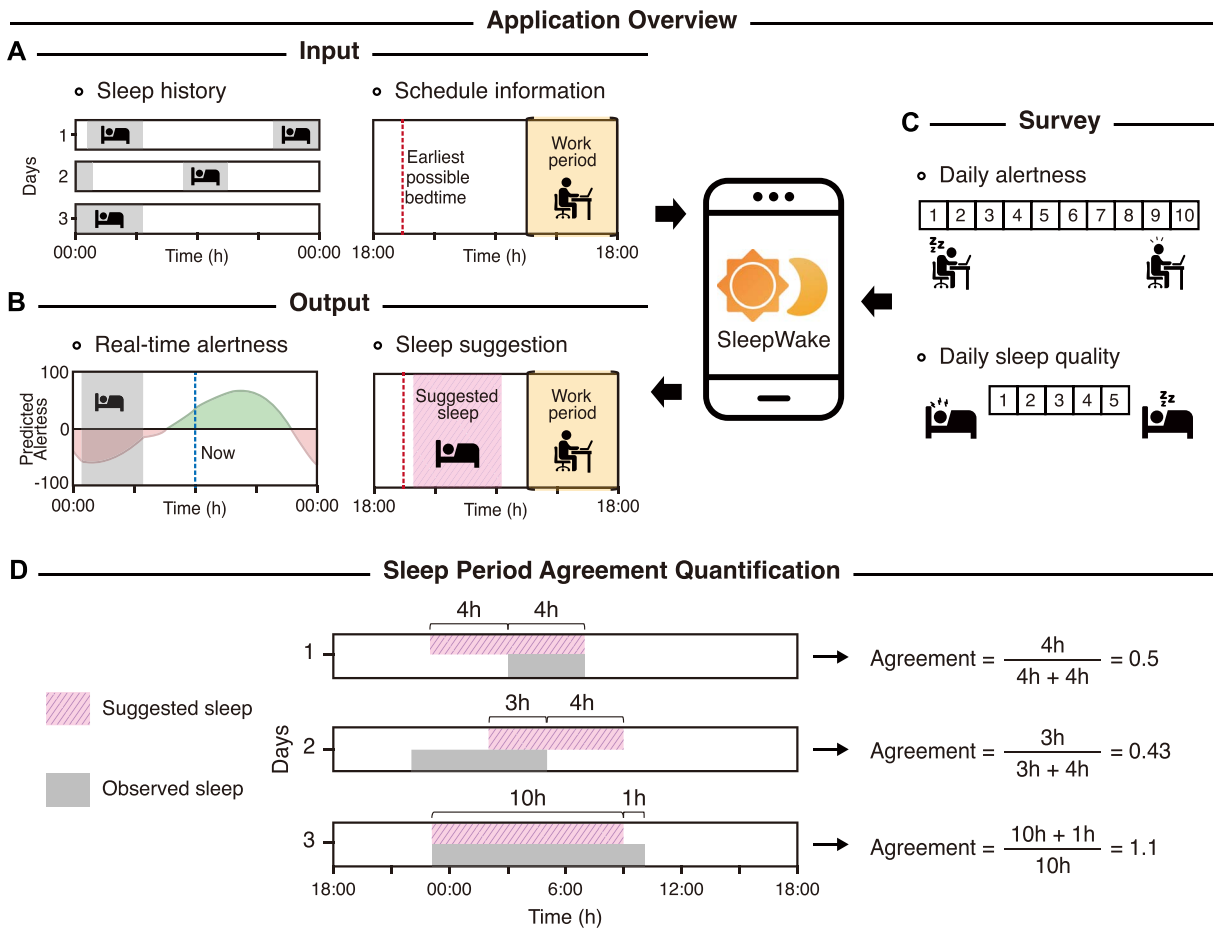

Figure 1. Method to evaluate the effectiveness of mathematical model-based sleep scheduling using the SleepWake app. (A) The SleepWake app processes users' sleep history (shaded regions, left), logged mainly via smartwatch or manually by the user if necessary. The app also collects users' schedule information, including the earliest possible bedtime for the current day (dashed line, right) and the designated work period for the following day (bracketed region, right). (B) Using these inputs, SleepWake computes and displays real-time alertness predictions (shaded region, left) around the current time (dashed line, left). The app also identifies optimal sleep patterns (hatched region, right) between the upcoming earliest possible bedtime (dashed line, right) and the work period (bracketed region, right) to enhance alertness during the work period, as determined by the embedded mathematical model. (C) SleepWake further collects user-reported assessments of overall alertness for the day (daily alertness, scale 1 ∼ 10), and the perceived quality of the previous night's sleep (daily sleep quality, scale 1 ∼ 5). (D) To evaluate user adherence to SleepWake's recommendations, the daily agreement between periods of observed sleep (shaded region) and SleepWake's recommended sleep (hatched region) was calculated as the proportion of their overlapping periods.

D, middle). If users fully adhered to the recommended sleep period, any additional sleep beyond the recommendation further increased the adherence score, allowing values to exceed one (Figure 1, D, bottom). This approach reflects the fact that the recommended sleep period represents the minimum duration needed to maintain a mean predicted alertness above zero. Therefore, sleeping beyond the recommendation—provided the full recommended window is included—may confer additional benefits.

Notably, equation (1) is a modification of the commonly used percentage overlap:

$$\text{percentage overlap} = \frac{\textit{Length}\left((\text{Observed Sleep}) \cap (\text{Recommended Sleep})\right)}{\textit{Length}\,(\text{Recommended Sleep})} \times 100\ (\%)\,, \tag{2}$$

which has also been used in previous studies to evaluate adherence to sleep recommendations [32]. To assess the robustness of our findings to the choice of adherence metric, we performed a sensitivity analysis by repeating all analyses using equation (2) in place of equation (1).

Mathematical model and sleep recommendation

The SleepWake app integrates a physiologically based mathematical model that simulates two key processes regulating human sleep and alertness: sleep pressure and the circadian process [29, 34]. Sleep pressure accumulates during wakefulness, increasing the need for sleep and reducing alertness, and dissipates during sleep to restore homeostasis, typically following an exponential decay trajectory. The circadian process represents fluctuations in the internal biological clock, influencing both alertness and sleep propensity. This process generally follows a sinusoidal pattern with a ∼24-h period, reaching its nadir near the minimum of core body temperature (CBTmin). Specifically, the circadian process is modeled using a Van der Pol oscillator, which allows for modulation by light exposure, resulting in variations in period, phase, and amplitude. Detailed model equations and parameter values are provided in the supplementary materials of our previous study [34].

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In the present work, we used the same parameters and initial conditions as in the previous study [34], with one exception: the intrinsic period of the circadian rhythm (τ_c) was set to 24.2 h, consistent with the original formulation [43]. Additionally, we adopted a previously used assumed light intensity profile for simulating the circadian process—250 lux during wakefulness and 0 lux during sleep, accounting for the effects of artificial lighting [34]. While light exposure was not directly measured, such an assumed light profile based on sleep patterns has been shown to accurately predict alertness and circadian phase shifts [34, 44–46].

In the model, the circadian process defines a dynamic sleep threshold. When sleep pressure exceeds this threshold, the individual is naturally predisposed to fall asleep. Consequently, remaining awake during these periods requires increased effort and is typically associated with reduced alertness. Conversely, when sleep pressure is below the sleep threshold, the individual is naturally inclined to stay awake, requiring minimal effort and corresponding to heightened alertness. Based on this framework, alertness at any given time is defined as the scaled difference between the sleep threshold and sleep pressure (i.e. sleep threshold − sleep pressure). This definition is consistent with our previous study, which successfully predicted the alertness levels of shift-working nurses in a real-world setting [29]. Specifically, the alertness level ($AL(t)$) at a given time t is defined as:

$$AL(t) = \mathbf{max}\left(\mathbf{min}\left(-100, \frac{100\left(H^{+}(t) - H(t)\right)}{3}\right), 100\right)$$

where $H^{+}(t)$ is the sleep threshold driven by the circadian process, and $H(t)$ is the value of sleep pressure. Here, the scaling factor 100/3 is used because $H^{+}(t) - H(t)$ typically ranges from −3 to 3. The max and min functions are applied to cap the resulting alertness value within the range [−100,100].

A previous study compared various sleep patterns using the physiologically based mathematical model and developed the ACSS pattern [29]. This pattern consists of two key components: a minimum necessary main sleep period and a late nap before work. The main sleep duration is determined based on the user's desired sleep onset and model simulations to ensure model-predicted alertness remains above zero at sleep offset. This split-sleep strategy is designed to sustain high model-predicted alertness during both work and nonwork periods, making it particularly beneficial for work schedules that deviate from traditional daylight hours.

Based on the ACSS pattern, SleepWake provides daily updated, personalized sleep recommendations that are more concretely defined and refined compared to the conceptual approach proposed in our previous study [29]. Recommendations are generated within a time window between the user's desired sleep onset and 1 h before the upcoming work onset. Here, the 1-h margin accounts for commute time and helps minimize the impact of sleep inertia, which can reduce alertness and typically dissipates within 30–60 min [47]. Importantly, SleepWake's recommendations dynamically incorporate the user's accumulating sleep history. As more data becomes available, the model re-simulates or extends predictions of sleep pressure and sleep threshold into subsequent days, building on prior simulation results. This allows SleepWake not only to incorporate past sleep history but also to continuously fine-tune the timing and duration of both main sleep and naps within the ACSS framework.

Specifically, SleepWake primarily recommends the minimum necessary main sleep duration required to reduce daytime sleepiness [34]. This main sleep begins at the user's desired sleep onset if the simulated sleep pressure exceeds the sleep threshold at that time, indicating that natural sleep is likely according to the model. The sleep continues until sleep pressure drops to meet the sleep threshold, ensuring that awakening does not require effort and alertness is above zero. If, however, the simulated sleep pressure is lower than the sleep threshold at the user's desired onset time, the main sleep is delayed until sleep pressure reaches the sleep threshold. If the main sleep period is insufficient to maintain the average predicted alertness above zero during future work periods, SleepWake suggests an additional late nap. The nap duration is set to the minimum necessary to ensure that the average predicted alertness remains above zero during work hours, and it extends until 1 h before the upcoming work onset. If there is no upcoming work period due to a scheduled day off, only the necessary main sleep period is recommended.

If the desired sleep onset and the next work shift are too close, allocating sufficient main sleep or maintaining alertness above zero during future work periods may be impossible, even after maximizing sleep duration. In such cases, SleepWake recommends advancing sleep onset while ensuring that the recommended sleep schedule avoids unfeasible periods, such as overlapping with work hours or times when alertness is above zero, which would make falling asleep physiologically difficult.

Data for retrospective analysis and reconstruction of SleepWake recommendations

To assess the effectiveness of the SleepWake intervention before the prospective clinical study, we utilized a dataset from a prior clinical study (Approval No. 2021-03-197) [29]. This dataset includes data from 71 female Korean nurses with over 2.5 years of shift work experience at a metropolitan hospital (≥2000 beds) in Seoul, Republic of Korea. The participants followed various shift schedules, including 8-h three-shift, 8-h two-shift, 12-h two-shift, and fixed night shifts. Between May 2021 and April 2022, data on the participants' sleep and work periods, along with KSS scores at the beginning and end of each shift, were collected using wrist actigraphy and daily sleep diaries.

With the collected sleep data, we reconstructed SleepWake's recommended sleep period—spanning from the desired sleep onset to the work period—by simulating the same mathematical model embedded in the SleepWake app. In this process, the desired sleep onset was gradually advanced until either (1) the average predicted alertness remained above zero during future work hours or (2) further advancement led to an unfeasible sleep schedule, mirroring the SleepWake app's approach of recommending earlier sleep onset in such cases (see *Mathematical model and sleep recommendation* section for details). We then compared the reconstructed recommended sleep periods with the observed sleep data and analyzed their association with the corresponding KSS scores. Notably, this analysis was not conducted in the previous study [29].

Participants in the prospective clinical study

This study involved 19 Emergency Medical Technicians (EMTs) working on a three-shift system at a private emergency medical service who volunteered to participate and were already using a smartwatch (Approval No. 2024-01-061). The mean age of the participants was 33.47 ± 7.90 years, and all participants were male (Table 1). The participants followed a 20-day rotating shift schedule consisting of the following cycle: 5 days of morning shifts, 2 days off, 5 days of night shifts, 2 days off, 5 days of evening shifts, and 1 day off. The shift times were as follows: morning (07:00 am

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Table 1. Demographics of the Participants

Characteristics	n (%)	M ± SD	Range
Male	19 (100)		
Age, y		33.47 ± 7.90	(26–53)
Height, cm		176.15 ± 5.71	(165–187)
Weight, kg		79.0 ± 8.49	(68–97)
Total career, y		8.75 ± 9.18	(1.17–29.42)
Shift work career, y		8.58 ± 8.94	(1.17–29.25)
Data collection period, days		31.58 ± 8.09	(23–45)

to 03:00 pm), evening (03:00 pm to 11:00 pm), and night (11:00 pm to 07:00 am). Data collection was conducted from February to May 2024.

Participants were instructed to wear their smartwatch while sleeping and to install the SleepWake mobile app on their smartphones. Among the participants, three used an Apple Watch, while the remainder used a Galaxy Watch. Data collection commenced on the first day of their day shift. Sleep data collected via the smartwatch was automatically synchronized with the SleepWake app. If sleep data were not recorded via the smartwatches, participants were asked to manually log their sleep in the app. During the baseline phase (first ∼6 days), the intervention-related features of the mobile app were locked to prevent participant access. On the first day of the intervention period, a password to unlock the app was provided, enabling participants to access the intervention content.

Participants were not required to strictly adhere to the sleep recommendations. This study design aimed to evaluate whether improvements in alertness occurred specifically among those who best adhered to the recommended sleep patterns during the intervention phase. Notably, this design is the key distinction from previous studies on sleep interventions using mathematical models [30–32].

TST, Midsleep, CBTmin, and phase similarity calculation

To assess the impact of sleep history on daily agreement and alertness, we calculated TST and midsleep for each day, based on the identified main sleep episode. The main sleep was defined as the longest sleep between shifts or the sleep episode immediately following a night shift. On days off without scheduled shifts, we applied a notional shift boundary from 07:00 pm to 07:00 pm (i.e. zero duration). This choice was based on the distribution of sleep onset times (Figure S1), which showed minimal sleep initiation around that time. Sleep episodes occurring between two main sleep episodes were classified as naps. TST was the sum of the main sleep and any subsequent naps, and midsleep was defined as the midpoint of the main sleep episode.

Additionally, to assess the effect of alignment between sleep timing and the optimal circadian phase for alertness, we estimated each participant's daily circadian phase—specifically, the timing of the CBTmin—using the mathematical model integrated within the SleepWake app. This model includes a variable (y; see Supplementary Information of [34] for details) that simulates the dynamics of the circadian pacemaker based on a Van der Pol oscillator and serves as a proxy for core body temperature [43], exhibiting a daily rhythm. The time at which this variable reaches its minimum during the daily cycle is identified as the CBTmin.

To quantify the alignment between two specific time points within the 24-h day—such as the average midsleep over the preceding 3 days and the midsleep on the current day, or the CBTmin and the current day's midsleep—we calculated their time difference in hours, referred to as the phase difference. Using this phase difference, phase similarity was calculated using the following equation:

$$\text{phase similarity} = \cos\left(\frac{2\pi\ (\text{phase difference})}{24h}\right).$$

Phase similarity ranges from −1 to 1, where 1 indicates completely aligned midsleep times, while −1 indicates a completely reversed sleep phase with a 12-h difference in midsleep times.

Statistical analysis

The associations of daily TST and agreement with mean KSS or alertness were explored by calculating Pearson correlation coefficients (*r*) and confidence intervals (CI) using the "corr.test" function in R. Additionally, partial correlations between daily agreement and alertness were also computed, controlling for the effects of TST. This was done using residuals from linear regression models for daily TST agreement and daily TST alertness. The linear regression models were fitted using the "lm" function in R, and correlations between the residuals, along with their CIs, were calculated with the "corr.test" function.

To evaluate the inter-individual effects of the intervention based on adherence, we categorized participants into high- and low-adherence groups and analyzed changes in average daily alertness, TST, and sleep quality from baseline (P1) to intervention (P2) phases. The significance of these changes was assessed using the two-sided Wilcoxon signed-rank test, conducted via the "stat_compare_means" function in R [48]. Additionally, the correlation between changes in the average daily agreement and alertness from P1 to P2 was examined by calculating *r* and CIs using the "corr.test" function in R. Based on the calculated *r*, we then conducted a post hoc power analysis to justify the study's sample size, using the "pwr.r.test" function in R.

To investigate the underlying mechanism of SleepWake's recommendations, we examined how daily TST, the mean TST over the preceding 3 days (TST_{prev}), and phase similarity between the average midsleep over the prior 3 days and the current day's midsleep relate to agreement and alertness. These relationships were assessed using multiple linear regression models fitted with the lm function in R. Additionally, we analyzed the association between alertness and either the phase similarity between the current day's midsleep and the predicted CBTmin, or the CBTmin itself, by calculating *r* and CIs using the "corr.test" function in R.

Results

Retrospective analysis for evaluation of SleepWake's effectiveness

To evaluate the effectiveness of SleepWake's sleep recommendations, we first conducted a retrospective analysis using data from a previous clinical study that involved 71 female shift workers (Figure 2, A; see *Materials and Methods* section for details) [29]. These participants were assigned to various rotating shift schedules, including two-, three-, or night-only shifts. They wore actigraphy devices and kept sleep diaries during a ∼14-day clinical experiment conducted in real-world settings. The study collected data on sleep and work periods, as well as subjective sleepiness levels, which were measured using the Karolinska Sleepiness

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Scale (KSS) score at the beginning and end of each work period (Figure 2, B).

Using these data, we reconstructed the estimated SleepWake recommendations (Figure 2, B; see *Materials and Methods* section for details). We then examined the association between the mean KSS score for each daily work period (mean KSS) and the agreement between the reconstructed SleepWake recommendations and observed sleep period (Figure 1, D) preceding the corresponding work period (Figure 2, C; see *Materials and Methods* section for details). Additionally, we analyzed the association between daily mean KSS and TST of the sleep period to assess whether TST acts as a confounding factor in the relationship between daily agreement and mean KSS (see *Materials and Methods* section for details).

The overall TST data exhibited an approximately normal distribution, with a mean of ∼470 min (Figure 2, D, top). When analyzing the full dataset, while daily TST did not correlate significantly with mean KSS ($r = -0.052$, $p = .298$), the daily agreement with the SleepWake recommendations exhibited a significant negative correlation with the mean KSS ($r = -0.230$, $p = 2.533e\text{-}06$) (Figure 2, E and F, top). This suggests that longer TST does not lead to greater alertness, whereas greater adherence to SleepWake's recommendations does lead to improved alertness. Furthermore, when restricting the analysis to data within narrower TST ranges centered around the mean (Figure 2, D, bottom), the correlation between TST and the mean KSS score remained insignificant ($r = -0.075$, $p = .232$), while the significant negative correlation between agreement and KSS score persisted ($r = -0.167$, $p = .007$) (Figure 2, E and F, bottom). These results indicate that even with similar TST, greater adherence to SleepWake recommendations can lead to higher levels of alertness. Notably, this significant association was also held when focusing exclusively on night shift days—periods typically associated with circadian disruption (Figure S2). Taken together, this retrospective analysis demonstrated the potential for adherence to SleepWake's recommendations to improve alertness in real-world settings, even under conditions of circadian misalignment such as night shift work.

Participants with high levels of adherence experienced improvements in alertness

Given the promising results of the retrospective analysis for SleepWake's evaluation, we conducted a prospective clinical study. We recruited 19 male shift workers (mean age: 33.47 ± 7.90 years) employed as EMTs who followed the same rotating three-shift schedule (morning, evening, night, and off) on a 20-day cycle (Figure 3, A). During the clinical study, participants were instructed to integrate the SleepWake app into their daily routines, ensuring that objective sleep data from their smartwatches were recorded in the app. In addition, participants logged daily ratings of alertness and sleep quality, while momentary alertness ratings were recorded at least three times per day. The logged data was transmitted to a connected server for collection.

The clinical study spanned ∼26 days, comprising a ∼6-day Baseline phase (P1) and a ∼20-day Intervention phase (P2) (Figure 3, B). During P1, the real-time alertness predictions and the recommended optimal sleep period for the following day (Figure 2, B) were not displayed to participants, although these outputs continued to be calculated in the background. In contrast, during P2, these features were fully visible and accessible to participants. Across all participants, P1 included only off days and a single shift type: 15 participants were assigned to the morning shift, while 4 participants were in the evening shift during this period.

Based on the collected data, SleepWake's recommendations were reconstructed using the same approach as in the retrospective analysis (see *Materials and Methods* section for details), and its daily agreement with the observed sleep period was calculated. Using the daily agreement changes between P1 and P2, we evaluated each participant's adherence. Specifically, we compared the average daily agreement over the entire P1 period (Figure 3, C, days 1–6) with a subset of P2 (Figure 3, C, days 21–26) that featured the same shift schedule and occurred 20 days later. Therefore, data from days 7 to 20 were excluded from this comparison, as the shift schedules during that time differed from those in P1. This approach enabled a fair assessment of adherence by controlling for shift type. Participants were then grouped based on individual changes in average daily agreement between P1 and P2: those with increased agreement in P2 were assigned to the high adherence (HA) group, while the remainder were categorized as the low adherence (LA) group (Figure 2, C). The HA and LA groups consisted of 10 and 9 participants, respectively. The average daily agreement increased by 21.8 ± 36.0%p from P1 to P2 in the HA group and decreased by 10.7 ± 6.3%p in the LA group.

For each group, we analyzed individual changes in daily alertness (Figure 1, C, top), TST, and sleep quality (Figure 1, C, bottom) between P1 and P2 (Figure 3, E–G). When comparing P1 and P2, the HA group exhibited a statistically significant increase in alertness ($p = .012$), rising from a mean of 4.67 to 5.64, whereas the LA group showed significant decreases ($p = .02$), from a mean of 5.64 to 4.49 (Figure 3, E). This indicates that alertness improved only for those who adhered closely to the recommendations. Notably, changes in average daily agreement and alertness were positively correlated (Figure S3; see *Materials and Methods* section for details), suggesting that greater adherence to the recommendations leads to greater improvements in alertness.

Interestingly, there was no significant change in TST in the HA group ($p = .058$), increasing from a mean of 356.0 to 397.3 min (Figure 3, F, left), suggesting that the improvement in alertness resulted from better alignment between sleep patterns and the recommended schedule rather than increased TST. The insignificant change in TST also indicates that adhering to the recommended schedule does not negatively impact TST. Furthermore, sleep quality remained stable in the HA group ($p = .72$), with a mean change from 2.93 to 3.05 (Figure 3, G, left), indicating that following the recommended sleep schedule does not have a significant adverse impact on sleep quality. In contrast, the LA group experienced a significant decrease in both TST ($p = .0039$), dropping from a mean of 374.6 to 332.7 min (Figure 3, F, right) and sleep quality ($p = .030$), decreasing from a mean of 3.19 to 2.63 (Figure 3, G, right), highlighting that deviation from the recommended sleep schedule is associated with reduced TST and sleep quality, which can negatively impact health and safety.

Taken together, the SleepWake app effectively identified optimal sleep periods to enhance alertness without compromising TST or quality, with significant benefits observed only when users actively follow the recommendations.

Greater agreement between observed and recommended sleep schedules is associated with greater daily alertness, even with similar TST

We demonstrated that greater adherence to the recommended sleep schedule over a period with consistent work patterns led to improvements in alertness (Figure 3). This finding suggests that alertness would generally be greater on days when users

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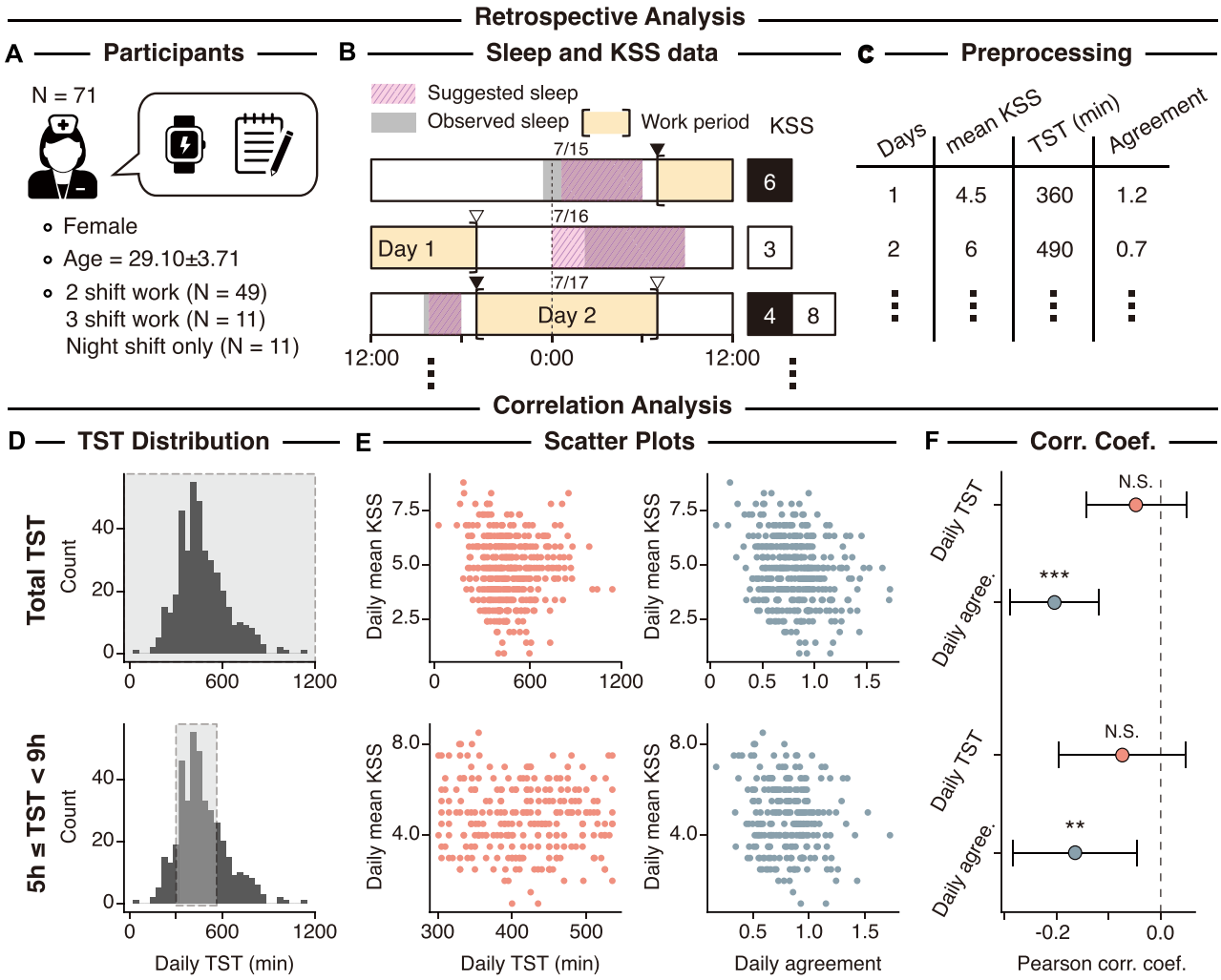

Figure 2. Greater agreement with SleepWake's recommendations is associated with greater daily alertness. (A) A retrospective analysis was conducted using data from 71 female shift workers (mean age = 29.10 ± 3.71) following three different rotating shift schedules. (B) The shift workers' sleep (shaded region) and work (bracketed region) period data, along with KSS scores at the beginning (filled triangle) and end (hollow triangle) of each work period, were collected via actigraphy and sleep diaries. Based on the collected data, the SleepWake recommendation (hatched region) was reconstructed. (C) For each daily work period, the mean KSS score, TST of the sleep period preceding the work period, and the agreement between the sleep period and the reconstructed SleepWake recommendations were calculated. (D–F) In the entire dataset (count = 407), where daily TST values followed an approximately normal distribution (D, top), daily TST (E, top left) did not show a significant correlation with the mean KSS (F, top). However, greater daily agreement (E, top right) showed a significant negative correlation with the mean KSS (F, top). This indicates that it is not greater TST but greater agreement with SleepWake recommendations that leads to a lower mean KSS (i.e. greater alertness). Error bars in (F) denote 95% CIs. Analysis within a narrower daily TST range—5 to 9 h (D, bottom, shaded region)—centered around the mean daily TST revealed no significant association between daily TST and mean KSS (E, bottom left; F, bottom). In contrast, the negative association between daily agreement and mean KSS remained significant (E, bottom right; F, bottom). This suggests that adherence to SleepWake recommendations contributes to lower mean KSS, even with similar TST. Here, N.S. $p > .05$, *$p < .05$, **$p < .01$, ***$p < .001$.

followed the recommended sleep more closely compared to days when they did not, as observed in the retrospective analysis (Figure 2). To test this hypothesis, we analyzed daily agreement and alertness data to assess whether a positive correlation existed between the two (Figure 4, A; see *Materials and Methods* section for detail). To account for the possibility that the improved alertness with increased agreement could be confounded by longer TST, we also examined the correlation between daily TST and alertness (Figure 4, A).

The overall TST data showed an approximately normal distribution centered around an average of 6 h (Figure 4, B, top). When analyzing the full dataset, both daily TST ($r = 0.343$, $p = 1.249e\text{-}12$) and agreement ($r = 0.423$, $p = 2.2e\text{-}16$) exhibited significant positive correlations with alertness (Figure 4, C and D, top). However, when restricting the analysis to data within narrow TST values clustered around the mean (Figure 4, B, mid and bottom), the correlation between TST and alertness disappeared ($r = -0.013$, $p = .879$ for mid; $r = 0.140$, $p = .148$ for bottom), while the significant positive correlation between agreement and alertness persisted ($r = 0.235$, $p = .004$ for mid; $r = 0.278$, $p = .003$ for bottom) (Figure 4, B–D, mid and bottom). These results indicate that, even with similar TST, alertness can vary depending on how closely sleep aligns with the recommended duration and timing.

To further examine this relationship, we assessed whether improved agreement consistently improves alertness, even after controlling for the effects of TST on agreement and alertness, using partial correlation [49]. Specifically, we trained linear regression models to quantify the relationships between TST and

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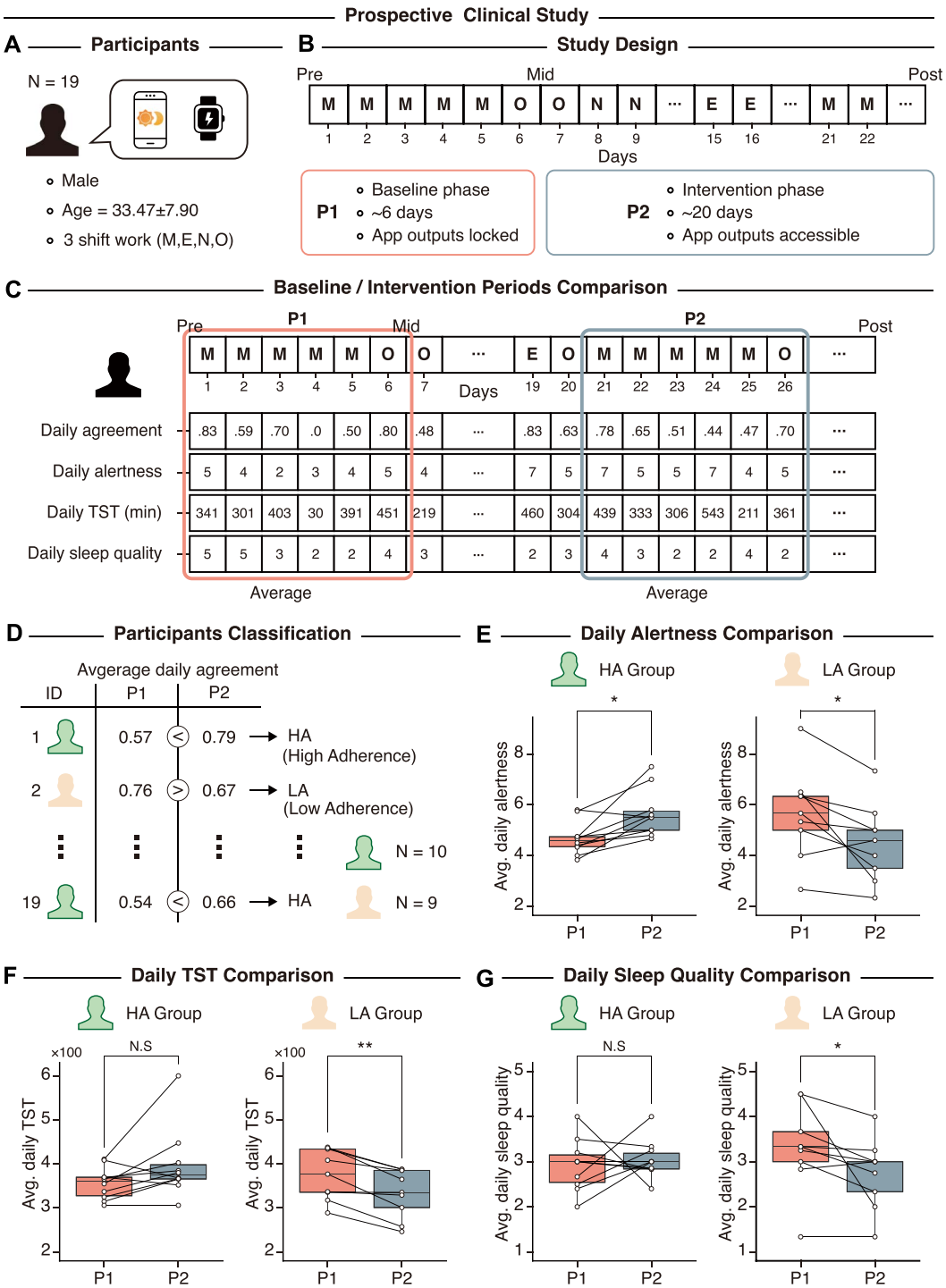

Figure 3. Participants with high adherence to SleepWake recommendations experienced improvements in alertness. (A) A clinical study to evaluate the effectiveness of SleepWake was conducted with 19 male shift workers (mean age = 33.47 ± 7.90) adhering to a rotating three-shift schedule (morning (M), evening (E), night (N), and off (O)). Participants utilized SleepWake and recorded sleep using smartwatches. (B) The study design included two phases: A baseline (P1; pre-mid) and an intervention (P2; mid-post) phase. During P1, spanning ~6 days, participants did not have access to SleepWake outputs, serving as a control condition. During P2, spanning, ~20 days, participants could access SleepWake's outputs. Comparisons of adherence and alertness across these phases were analyzed to assess the intervention's effectiveness. (C) Daily agreement, along with recorded daily alertness, TST, and sleep quality, were averaged across P1 (left box) and the corresponding days in P2 with the same shift types (right box) to ensure consistency in comparison. (D) Based on the average daily agreement in P1 and P2, participants were grouped: The high adherence (HA) group ($n = 10$) included those whose average daily agreement in P2 exceeded that in P1, while the low adherence (LA) group ($n = 9$) included those whose daily agreement did not increase. (E) The HA group exhibited a significant daily alertness increment during P2 compared to P1, whereas the LA group showed a significant decline in P2 relative to P1. (F and G) In the HA group, changes in daily TST (F) and sleep quality (G) were insignificant. In contrast, the LA group exhibited significant reductions in both measures during P2. These results demonstrate that adherence to the recommended sleep schedule enhances alertness while preserving TST and sleep quality, whereas deviations from the schedule are associated with reductions in both TST and sleep quality. Here, N.S. $p > .05$, $^{*}p < .05$, $^{**}p < .01$, $^{***}p < .001$.

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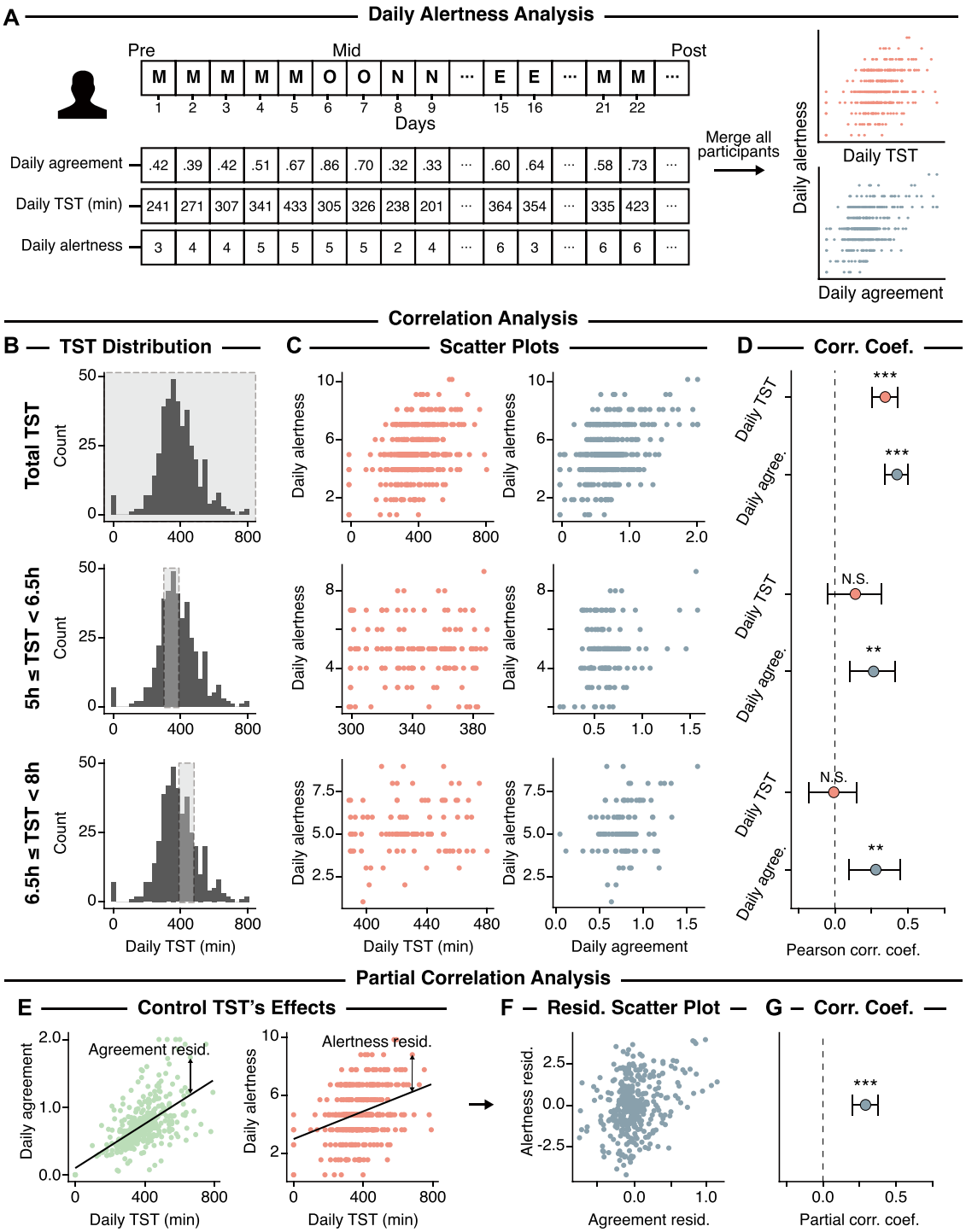

Figure 4. Greater agreement with SleepWake's recommendations is associated with greater daily alertness, even with similar TST. (A) To examine the daily effects of TST and agreement on alertness, their correlations were analyzed using aggregated participant data. (B–D) In the entire dataset (count = 405), where daily TST values followed a normal distribution (B, top), daily TST and alertness (C, top left) showed significant correlation (D, top). Daily agreement and alertness (C, top right) also showed a significant correlation (D, top). This indicates that greater agreement and longer TST are each associated with greater daily alertness. Error bars in (D) denote 95% CIs. Analysis within specific narrow daily TST ranges—5 to 6.5 h (B, middle, shaded region) and 6.5 to 8 h (B, bottom, shaded region)—centered around the mean daily TST, revealed no significant association between daily TST and alertness (C, middle and bottom left; D, middle and bottom). However, the positive correlation between daily agreement and alertness remained significant (C, middle and bottom right; D, middle and bottom), suggesting that adherence to SleepWake recommendations contributes to greater alertness, even with similar daily TST. (E and F) To examine the relationship between daily agreement and alertness while accounting for the effects of TST, their partial correlation was calculated using residuals from linear regression models between daily TST agreement and daily TST alertness. (G) A positive partial correlation was observed, indicating that adherence to the suggested sleep schedule enhances daily alertness, independent of TST. Here, N.S. $p > .05$, *$p < .05$, **$p < .01$, ***$p < .001$.

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agreement, as well as between TST and alertness (Figure 4, E). To control for the influence of TST, we calculated the residuals of these linear regression models. The correlation between the residuals (Figure 4, F) represents the partial correlation between agreement and alertness after removing the effect of TST [50]. The resulting partial correlation was significantly positive ($r = 0.290$, $p = 2.834e\text{-}09$) (Figure 4, G), demonstrating that greater agreement is associated with greater alertness even when the effect of the length of the TST is controlled. Notably, this positive association remained robust when the analysis was limited to night shift days and subsequent off days—periods typically marked by circadian disruption (Figure S4). These findings suggest that the relationship between agreement and alertness holds even under conditions of circadian misalignment caused by night shifts.

Taken together, adherence to SleepWake's sleep recommendations appears to enhance daily alertness, even when daily TST is similar. These findings emphasize that optimizing alertness requires not only adequate TST but also precise alignment of sleep timing with personalized sleep recommendations.

Sleep periods should be adjusted according to previous TST and circadian rhythms to enhance alertness

Through the previous analyses, we confirmed that even when TST is identical, closer alignment with the recommended sleep schedule results in greater alertness (Figures 2 and 4). This raises the question: what variations in sleep history affect agreement and thus alertness despite similar TST? To investigate this, we analyzed two distinct sleep history periods for one participant over 4 days, both featuring similar TST on the final day but differing in agreement and resulting alertness.

For one participant (Figure 5, A), we found that in one period (Figure 5, A, left), insufficient average TST over the preceding 3 days ($TST_{prev} = \sim 6$ h) led to a longer recommended sleep period to maintain high alertness. Conversely, when TST_{prev} was sufficient ($TST_{prev} = \sim 9$ h) (Figure 5, A, right), the recommended sleep period was shorter. For the longer recommended sleep period caused by insufficient sleep in the preceding 3 days, TST (~6 h) on the final day was insufficient (Figure 5, A, left) resulting in lower alertness. In contrast, the same TST (~6 h) was sufficient for the shorter recommended sleep, resulting in greater alertness.

For another participant (Figure 5, B), TST_{prev} was similar across two histories (~355 min), but the timing of the previous sleep periods differed. In one case (Figure 5, C, left), the average midsleep time of the previous sleep periods (~11:49) was significantly delayed compared to the midsleep time of the final day (~2:34), resulting in a large phase difference (~9.26 h) and low phase similarity ($\cos\left(\frac{2\pi\,(\text{phase difference})}{24\ \text{h}}\right) \approx -0.75$) (see *Materials and Methods* section for details on midsleep and phase similarity). Such delayed previous sleep history caused a delayed circadian phase that had previously been entrained to past sleep patterns. Then, the recommended sleep period, adjusted to align with the delayed circadian rhythm, was also delayed. Consequently, even with a long TST (>9 h), misalignment with the recommended sleep period resulted in low agreement and alertness. Conversely, in the other history (Figure 5, B, right), although the final day's TST was shorter (<9 h), the phase of the prior sleep periods was better aligned (~0.74). This alignment allowed the observed sleep period to nearly match the recommended sleep period, resulting in greater agreement and alertness.

The findings from the case analyses, where greater TST_{prev} and phase similarity led to greater agreement and alertness with similar TST level, were further examined using the daily data (Figure 5, C and D). When analyzing the data within narrow ranges of daily TST (5–6.5 h in Figure 5, C and 6.5–8 h in Figure 5, D), we observed a trend that increasing daily TST_{prev} and phase similarity was associated with greater agreement and alertness. To investigate this relationship statistically, we conducted multiple linear regression analyses, predicting agreement or alertness based on current TST, TST_{prev}, and phase similarity (Tables 2 and 3). All variables were significant contributors, with increases in each leading to improvements in agreement and alertness. This suggests that even after accounting for the effects of current TST, greater TST_{prev} and phase similarity still result in increased agreement and alertness.

Taken together, we established a key principle for sleep interventions aimed at enhancing alertness: when prior TST is insufficient, longer sleep should be recommended (Figure 5, E), and the phase of the recommended sleep period must closely align with prior sleep history (Figure 5, F). These findings highlight the limitations of generic, one-size-fits-all sleep interventions, demonstrating that improving alertness requires individualized sleep patterns that account for both past sleep duration and phase. This also emphasizes the importance of mobile apps like SleepWake, which provide personalized daily sleep schedules by continuously integrating updated sleep history to optimize alertness effectively.

Discussion

This study provides the first clear evidence that mathematical model-driven sleep recommendations can enhance alertness. We integrated a previously developed model-based sleep recommendation framework into a mobile app for real-world application (Figure 1). A retrospective analysis of data from a previous clinical study involving 71 female shift workers demonstrated that precise adherence to model-based sleep recommendations can lead to improved alertness (Figure 2). In a subsequent clinical study with 19 male shift workers using this app, only those who more precisely adhered to the recommended sleep schedules experienced significant improvements in alertness without compromising TST or sleep quality (Figure 3 and Figure S3). Furthermore, on days with greater adherence to the recommendations, alertness was enhanced, even when TST was similar (Figure 4). Finally, a detailed analysis of cases with similar TST but differing alertness levels, alongside multiple linear regression incorporating the duration and phase of the previous sleep, highlighted a key difference of our model that leads to enhanced alertness over conventional one-size-fits-all approaches: the model dynamically recommended supplemental sleep when prior sleep was insufficient and aligned sleep phases with individual historical patterns (Figure 5).

In the retrospective analysis, alertness was measured using the KSS at the beginning and end of each work shift, thereby capturing momentary sleepiness levels. In contrast, the prospective study asked participants to evaluate their overall alertness across the day (see *Materials and Methods* section for details). This discrepancy in measurement scales and timing may partly explain why TST was not significantly associated with alertness in the retrospective analysis (Figure 2) but showed a significant association with daily alertness ratings in the prospective study (Figure 4). Notably, despite these differences in the alertness items, we observed consistent associations between the alignment of observed sleep with SleepWake's recommended sleep and both alertness measures—even when analyses were restricted to a narrow range of TST

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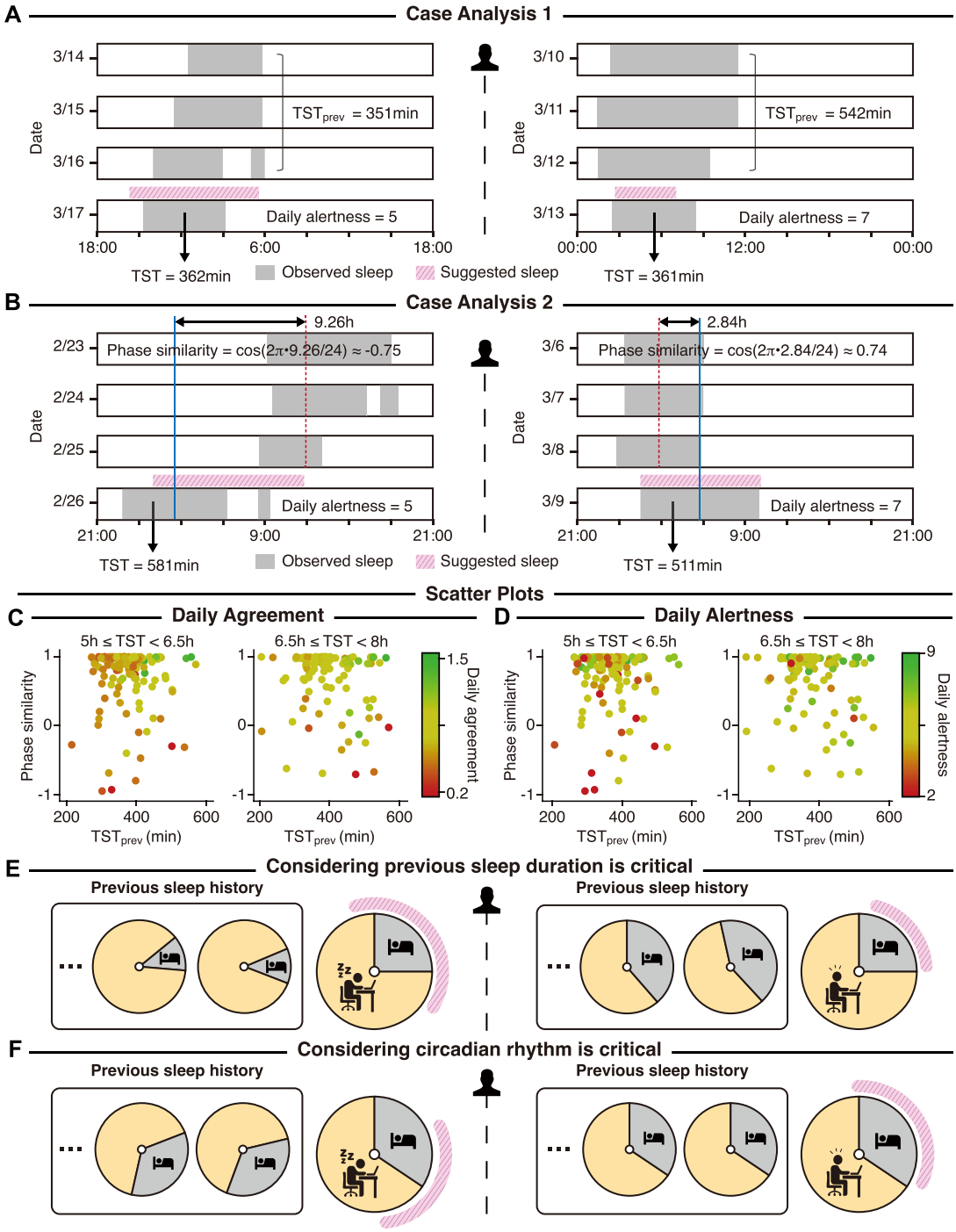

Figure 5. Sleep periods should be adjusted based on previous TST and circadian rhythm to promote high alertness. (A) Representative sleep histories (shaded region) over 4 days, with similar TST on the final day, and their corresponding recommended sleep periods (hatched region) were analyzed. For the same participant, a sleep period (TST = 362 min) following a shorter average TST over the previous 3 days (TST_{prev} = 351 min) failed to meet the suggested sleep requirement, resulting in low alertness (daily alertness = 5; left). In contrast, a sleep period (TST = 361 min) following a longer TST_{prev} (TST_{prev} = 542 min) aligned with the suggested sleep period, leading to greater alertness (daily alertness = 7; right). (B) For another participant, a long sleep period (TST = 581 min) with a midsleep time (solid vertical line) that was largely misaligned with the average midsleep over the previous 3 days (dashed vertical line) (phase similarity = −0.75) did not meet the suggested sleep requirement, resulting in low alertness (daily alertness = 5; left). Conversely, a shorter sleep period (TST = 511 min) with a midsleep time that was more closely aligned with the prior average midsleep time (phase similarity = 0.74) nearly satisfied the suggested sleep requirement, leading to high alertness (daily alertness = 7; right). (C and D) Scatter plots of daily TST_{prev} and phase similarity, with colors representing daily agreement (C) and alertness (D), for daily TST ranges of 5 to 6.5 h (left) and 6.5 to 8 h (right). Within these narrow daily TST ranges, greater TST_{prev} and phase similarity appear to be associated with greater agreement and alertness. (E and F) These results demonstrate that incorporating previous sleep history is critical for improving alertness: The recommended sleep period needs to be lengthened when the prior TST is short, and sleep periods must be closely aligned with prior sleep phases to optimize alertness.

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Table 2. Multiple Linear Regression Analysis for Daily Agreement

Variables	β	t	P-values	SE
Intercept	−0.4995	−6.509	<.0001	0.0767
TST	0.0014	12.656	<.0001	0.0001
TST_{prev}	0.0015	8.693	<.0001	0.0002
Phase similarity	0.1603	5.639	<.0001	0.0284

β, regression coefficient; *t*, *t*-statistics; *p*, *p*-value; *SE*, standard error; TST, total sleep time.

Table 3. Multiple Linear Regression Analysis for Daily Alertness

Variables	β	t	P-values	SE
Intercept	−1.7120	3.714	.0002	0.4610
TST	0.0043	6.445	<.0001	0.0007
TST_{prev}	0.0032	3.139	.0018	0.0010
Phase similarity	0.7244	4.241	<.0001	0.1708

β, regression coefficient; *t*, *t*-statistics; *p*, *p*-value; *SE*, standard error; TST, total sleep time.

(Figures 2 and 4). This suggests that the SleepWake recommendation may enhance not only immediate alertness during work shifts but also participants' broader perception of daily alertness, reinforcing the potential benefit of personalized sleep scheduling.

We found that recommended sleep timing aligned more closely with observed sleep and was associated with improved daily alertness, particularly when sleep timing remained consistent with prior history—a proxy for stable circadian alignment (Figure 5, B–D). This outcome stems from the model underlying the SleepWake app, which estimates each user's sleep threshold and suggests sleep only when the modeled homeostatic sleep pressure exceeds this threshold. By design, this approach encourages sleep during biologically favorable phases—typically near the circadian nadir, which closely corresponds to CBTmin—a time conducive to both sleep initiation and maintenance [51]. Further supporting this interpretation, we observed that days with greater alignment between the estimated CBTmin and the midpoint of observed sleep were associated with higher levels of alertness upon waking (Figure S5; see *Materials and Methods* section for details on CBT min estimation). In contrast, absolute phase shifts in CBTmin (e.g. advances or delays) were not significantly associated with changes in alertness. These findings suggest that maintaining alignment with one's circadian rhythm—rather than merely avoiding phase shifts—is critical for sustaining daytime alertness for shift workers.

In this study, we used a modified metric (equation 1) to evaluate the agreement between the recommended and observed sleep that extends the standard percentage overlap (equation 2) by allowing values greater than one when the observed sleep duration fully encompasses the recommended sleep window. This modification reflects the interpretation of the recommended period as a minimum threshold necessary for maintaining alertness, suggesting that additional sleep may provide further benefits and therefore should not be penalized. However, this approach may lead to inflated agreement scores when the observed sleep duration substantially exceeds the recommendation. To address this concern, we performed a sensitivity analysis using the conventional percentage overlap metric, which caps agreement values at one (Figure S6). The results remained qualitatively consistent, supporting the validity of our conclusions regardless of the specific adherence metric used.

One of the assumptions used in our model simulations was a light intensity of 250 lux during wakefulness and 0 lux during sleep. While this light profile has been commonly used in previous studies [34, 44–46], it may overestimate nighttime light exposure during wakefulness, which is typically lower and varies depending on occupation and lighting environment. To assess the sensitivity of our model to this assumption, we conducted an additional analysis using a modified light schedule in which light intensity was set to 40 lux between 08:00 pm and 08:00 am, reflecting reduced ambient light during typical nighttime hours and aligning with previous modeling studies (Figure S7) [29, 35]. The results under this adjusted light scheme did not result in any qualitative changes in the recommended sleep schedules or overall model behavior. These findings suggest that the core functionality and recommendations of SleepWake are robust to moderate variations in light intensity assumptions.

In the LA group of the prospective study, we observed a significant decline in sleep quality, suggesting that sleeping outside the recommended sleep window may negatively impact sleep quality (Figure 3). However, in the HA group, no significant improvement in sleep quality was observed. This may be due to the limited sample size, which could have reduced statistical power. Alternatively, it suggests that changes in sleep patterns alone may not be sufficient to effectively enhance sleep quality, underscoring the need for additional interventions. For instance, previous studies employing mathematical models not only recommended optimal sleep schedules but also incorporated education on lifestyle modifications for shift workers, leading to significant improvements in perceived sleep quality [30, 31]. Integrating our sleep intervention framework with such strategies could potentially enhance intervention effectiveness.

Another promising intervention that could have synergistic effects when combined with our approach is controlled light exposure. Light is a primary regulator of circadian timing, influencing alertness both indirectly through circadian entrainment [52] and directly via its acute alerting effects [53]. As a result, regulating light—its intensity, timing, or duration—serves as an effective tool for helping shift workers adapt to night shifts and enhance alertness [54–57]. Notably, optimizing light exposure protocols to enhance their effectiveness has been explored using mathematical models that predict circadian rhythm responses to light [58–61]. Our model includes a component that simulates circadian adjustments based on light exposure (see *Materials and Methods* section for details), making it well-suited for integration with light-based interventions. Future research could investigate how these approaches can be combined to maximize their effectiveness in promoting alertness and circadian adaptation.

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While the current research has primarily focused on alertness data to evaluate the effectiveness of interventions, it is also essential to examine other sleep-related indicators. For instance, assessing whether the proposed sleep timing helps individuals fall asleep more easily compared to their previous patterns is crucial, as explored in previous studies [30–32]. Thus, analyzing whether SleepWake recommendations lead to improvements in the insomnia severity index or reductions in sleep onset latency would be critical. Furthermore, mood is also well-known to be closely linked to sleep and circadian rhythms [44, 45, 62], making it another critical area for exploration. A compelling direction for future research would be to examine whether our proposed sleep recommendations are effective in enhancing mood. Additionally, it would be valuable to determine if mood improvements require different sleep patterns than those optimized for improving alertness.

Our sleep recommendations guide users to wake up at least 1 h before their scheduled work shift, accounting for the effects of sleep inertia (see *Materials and Methods* section for details). However, the duration of sleep inertia can vary depending on prior sleep history and wake-up time, with some individuals experiencing little to no sleep inertia while others experience it for over an hour [47, 63]. Notably, recent research has developed models to predict the effects and duration of sleep inertia on alertness [64]. By incorporating insights from these models into our framework, we could move beyond rigid recommendations of waking up exactly 1 h before a shift. This would allow for more flexible and potentially more effective sleep recommendations.

The mathematical model we currently use assumes uniform parameters for all individuals. While interventions based on this model have successfully improved alertness, evidence suggests significant physiological differences between individuals, highlighting the need for personalized model parameters to enhance effectiveness. For example, long sleepers and short sleepers may differ in the time scales of sleep pressure accumulation and dissipation [65]. Additionally, studies have shown substantial variability in individuals' sensitivity to light, a key factor in regulating circadian rhythms [66]. Recent research has demonstrated that optimizing personalized parameters in a mathematical model can better align model-predicted sleep patterns with observed sleep behaviors [67]. Similarly, by tailoring our model, we could estimate personalized parameters that more accurately reflect individual differences and predict changes in alertness based on sleep history. This work might require identifying the conditions under which these parameters can be reliably estimated.

Despite promising findings, several limitations exist. First, the prospective clinical study included only 19 participants. Since the prospective clinical study was designed as an exploratory pilot to assess the feasibility and potential efficacy of the SleepWake app, a priori sample size calculation was unfeasible. The decision to recruit 19 workers was driven primarily by logistical constraints, including recruitment feasibility and workforce availability. Notably, a post hoc analysis of a key finding—specifically, the correlation between average change in agreement with SleepWake recommendations and average change in alertness from P1 to P2 (Figure S3)—suggested that the study had 85 per cent statistical power (see *Materials and Methods* section for details), indicating that it was adequately powered to detect the observed effect. However, assuming a more conservative effect size in future studies would require a larger sample size to ensure sufficient statistical power. This suggests that additional studies with more participants may be necessary to replicate and validate the findings.

Second, as EMTs, participants may have had rest opportunities during nonemergency periods, potentially reducing fatigue. Furthermore, while participants followed a three-shift system, their consistent 20-day rotation pattern may have influenced the outcomes. Notably, a retrospective analysis in this study was conducted using data from 71 shift workers with different genders, work types, and shift schedules (see Figure 2 and *Materials and Methods* section for details). This analysis supports the generalizability and broader applicability of our findings, suggesting that the promising results observed in the prospective clinical study could extend to a wider range of occupational groups and shift schedules. It is also essential to explore whether this intervention benefits nonshift workers.

Third, this study primarily relied on subjective measures of alertness and sleep quality. For more robust validation, future studies should incorporate objective assessments. For example, integrating tests such as the Psychomotor Vigilance Task into the app could provide an objective measure of alertness [68]. Furthermore, accurately predicting sleep stages may be essential for evaluating self-reported sleep quality more effectively, which could be enhanced by recent advancements in precise sleep–wake state and sleep stage prediction algorithms [69–73].

Finally, due to structural constraints in our app, we could not determine whether the sleep records originated from wearable device sensors, manual input by participants, or automated sleep detection algorithms (e.g. time-in-bed estimates based on phone or watch usage patterns). As a result, all sleep entries were uniformly treated as observed sleep data, regardless of origin. This lack of source verification may have introduced variability in data accuracy, potentially affecting the calculation of individual adherence scores and their association with alertness outcomes. For instance, automated estimations may overestimate actual sleep duration, while manually entered data are susceptible to recall bias. Future work should focus on integrating data pipelines that explicitly identify the source and method of sleep detection, enabling more precise analyses and interpretations.

Supplementary material

Supplementary material is available at *SLEEP* online.

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Author contributions

EYJ and JKK conceptualized the research. YMS and JKK developed the methodology. SJC and HRP collected the experimental data. YMS analyzed the data. YMS, HJJ, and RHW developed the mobile application. EYJ and JKK supervised the project. YMS and CSJ wrote the draft of the manuscript. All authors contributed valuable feedback, refining the manuscript for intellectual content.

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Disclosure statement

Financial disclosure: None declared.

Non-financial disclosure: The authors declare the following competing interests: The mobile application SleepWake was developed by the authors, and there are plans for its public release. Additionally, a patent has been filed related to the underlying technology of SleepWake. These interests could be perceived as potential conflicts of interest.

Data availability

The data underlying this article will be shared on reasonable request to the corresponding author.

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